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F.2.4 Sample Exam Answers and Videos Questions 5-6

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Question 5

0 points possible (ungraded)

5. Consider the vectors $a_0 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ and $a_1 = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$.

(a) (6 points) Compute orthonormal vectors q_0 and q_1 so that q_0 and q_1 span the same space as a_0 and a_1 .

(b) (4 points) Let $A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 2 \end{pmatrix}$. Give Q and R such that $A = QR$ and R is upper triangular.

- $\rho_{0,0} = \|a_0\|_2 = \sqrt{a_0^T a_0} = \sqrt{2}$.

- $q_0 = a_0/\rho_{0,0} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} / \sqrt{2} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}.$

- $\rho_{0,1} = q_0^T a_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}^T \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} = \frac{1}{\sqrt{2}}$

- $a_1^\perp = a_1 - \rho_{0,1}q_0 = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} - \frac{1}{\sqrt{2}}\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} - \frac{1}{2}\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1/2 \\ -1/2 \\ 2 \end{pmatrix}.$

- $\rho_{1,1} = \|a_1^\perp\|_2 = \sqrt{a_1^{\perp T} a_1^\perp} = \sqrt{1/4 + 1/4 + 4} = \sqrt{9/2} = \frac{3}{\sqrt{2}}.$

- $q_1 = a_1^\perp / \rho_{1,1} = \frac{\sqrt{2}}{3} \begin{pmatrix} 1/2 \\ -1/2 \\ 2 \end{pmatrix}$

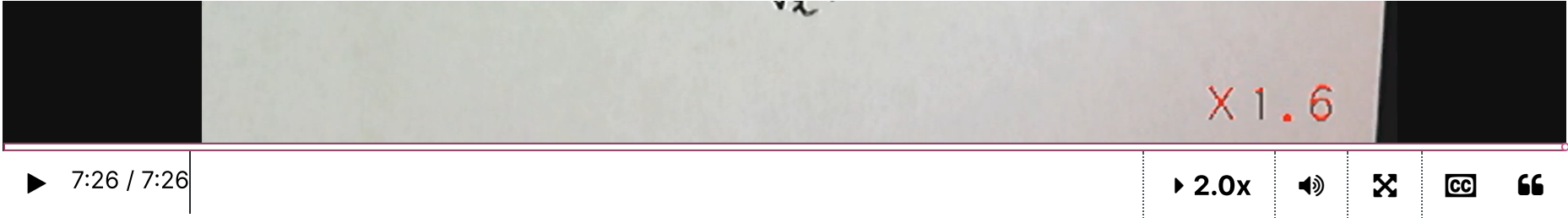
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i Answers are displayed within the problem

F.2.4 Sample Exam Answers and Videos Questions 5-6

Question 5

$$\begin{aligned} r_{01} &= q_0 \cdot a_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 2 \end{pmatrix} = \frac{1}{\sqrt{2}} \\ a_1^\perp &= a_1 - r_{01} \cdot q_0 = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} - \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 2 \end{pmatrix} \\ p_{11} &= \|a_1^\perp\| = \sqrt{\frac{1}{4} + \frac{1}{4} + 4} = \sqrt{\frac{18}{4}} = \frac{3}{\sqrt{2}} \\ q_1 &= \frac{1}{p_{11}} a_1^\perp \\ &= \frac{\sqrt{2}}{3} \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 2 \end{pmatrix} \end{aligned}$$



Video

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Question 6

0 points possible (ungraded)

6. Consider the matrix $A = \begin{pmatrix} 1 & -1 & -2 & 1 \\ -1 & 3 & 2 & 0 \end{pmatrix}$.

- (a) (3 points) Give a basis for the column space of A .
- (b) (3 points) Give a basis for the row space of A .
- (c) (6 points) Find two linearly independent vectors in the null space of A .

6. Consider the matrix $A = \begin{pmatrix} 1 & -1 & -2 & 1 \\ -1 & 3 & 2 & 0 \end{pmatrix}$.

I am not giving answers here. I'm tired...

- (a) (3 points) Give a basis for the column space of A .
- (b) (3 points) Give a basis for the row space of A .
- (c) (6 points) Find two linearly independent vectors in the null space of A .

Additional comment for Part a) since the video does not quite use the process described in the text and other videos.

Ordinarily, one goes back to the ORIGINAL matrix in order to identify the columns that form a basis for the column space of the matrix. The reason has to do with the fact that in the process of transforming the matrix to row echelon form, rows may be swapped (pivoted) which would change the span of the column in the row echolon form wrt the span of the original columns.

In this particular case, two things can be noted:

1) The column space of the matrix is clearly all of $\mathbb{R}^2$ and hence any vectors that can be easily identified as being linearly independent form a basis. The two columns that Maggie picks from the row echolon form clearly have the property that they are linearly independent because of the zero pattern in them; and

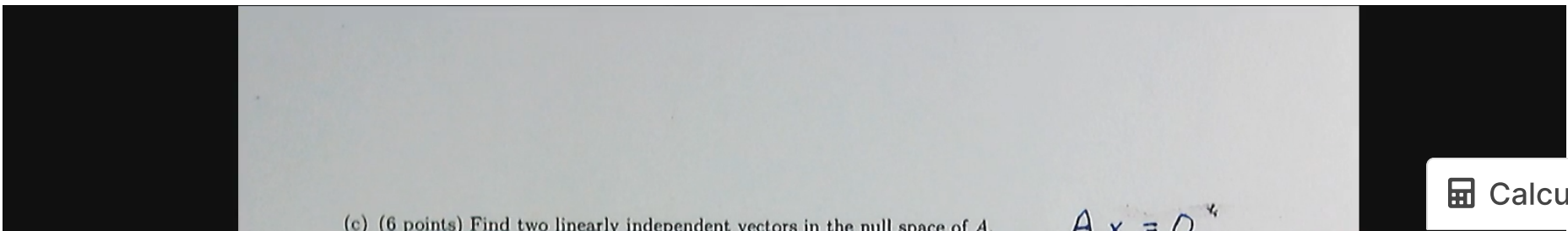
2) No pivoting happened.

Robert promised one of the participants that he would add an explanation of why the columns in the original matrix form a basis if one looks at where the pivots appear in the row echolon form. It is time to do so, now that things are quieting down a bit!

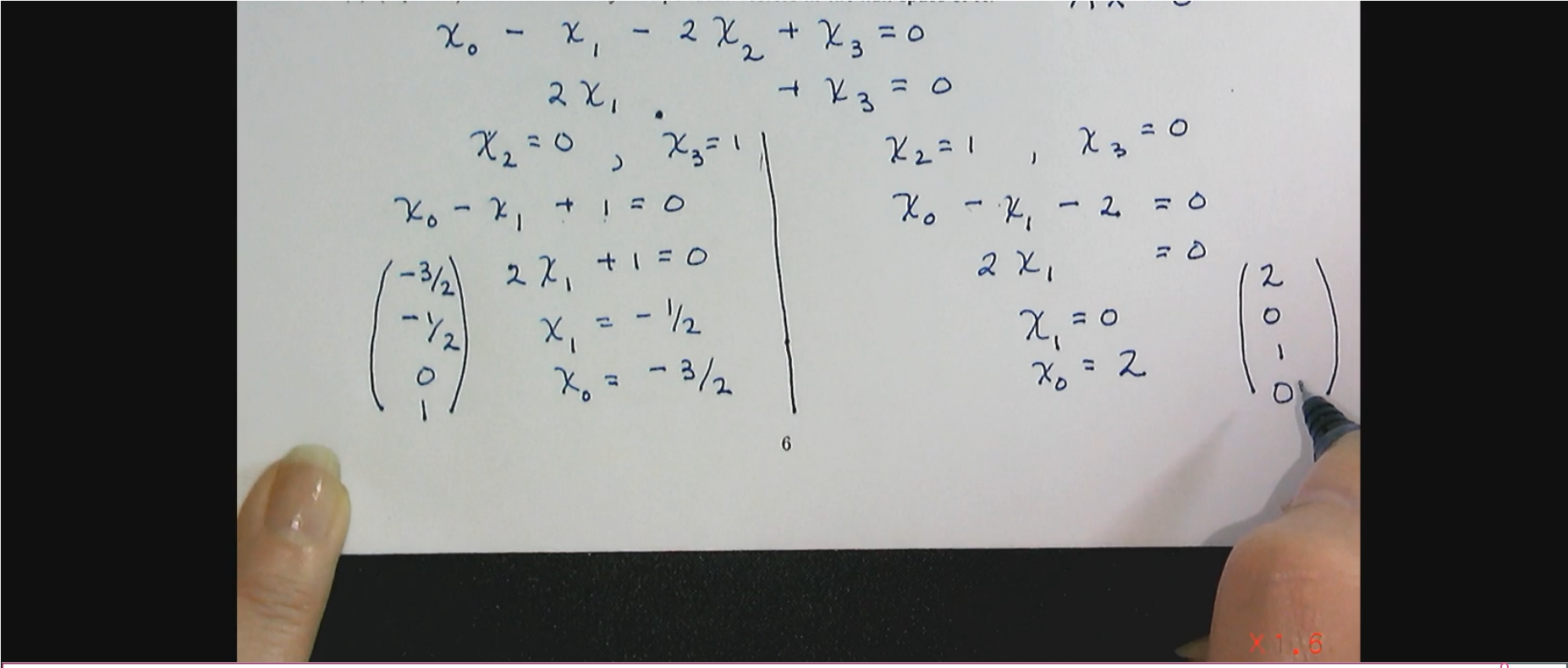
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Question 6



Calculator



6:52 / 7:07

Video

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